TEXORA · LECTURE SHEET

# Double Cloth

*Two complete cloths woven one above the other in the same shed sequence — two warps, two wefts — and joined into a single fabric by stitching points built into the design. Double cloth gives thickness, insulation, two different faces and, in the tubular form, seamless sacks and hoses. Its design grid has strict rules, and this sheet states and verifies them.*

TE-3205 · FABRIC STRUCTURE AND DESIGN · TOPIC 7 · LEVEL 3 TERM 2

## 1. The system of threads

A double cloth has **four yarn systems**: face warp and face weft, back warp and back weft. In the design they are interleaved in a fixed **arrangement**, most commonly **1 face : 1 back** in both directions: ends run F, B, F, B… and picks are inserted f, b, f, b… (lowercase for back picks in this sheet's figure labels). Each layer weaves its own weave (here both layers plain) in its own threads; the two layers are then **stitched** — joined at chosen points — or left unstitched where a tube or pocket is wanted. Design paper records all four systems on one grid under three rules:

- **On a face pick:** mark the face ends according to the face weave. The back ends stay **down** (blank) so the face pick passes *above* the whole back layer — except at stitching points (Rule 3).
- **On a back pick:** mark **all face ends up** (the face layer must be lifted clear so the back pick passes *below* it), and mark the back ends according to the back weave.
- **Stitching** is added as extra marks: either a **back end is raised over a face pick** (back-warp stitching — the stitch shows, if at all, on the back) or a **face end is lowered under a back pick** (a blank cut into the all-up row — face-warp stitching). The back-warp method is preferred when the face must stay perfectly clean, and is the one drawn in Figure 1.

## 2. The design, verified row by row

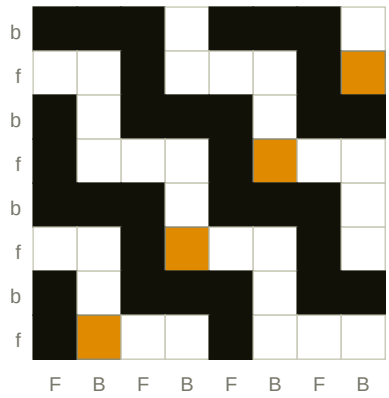



Figure 1 — Double plain cloth, arrangement 1F:1B in warp and weft, repeat 8 ends × 8 picks (4 face + 4 back threads each way). Columns are labelled F/B (face/back ends), rows f/b (face/back picks). Orange squares are the stitching points: back end raised over a face pick, one stitch per face pick, distributed one to each back end in turn. Verify the rules: every face-pick row carries exactly 2 face marks (plain weave) + 1 stitch = 3 marks; every back-pick row carries 4 face marks (all face ends lifted) + 2 back marks (back plain weave) = 6 marks.

Read one row aloud to fix the method. **Pick 3 (a face pick, third row from the bottom)**: face ends F2 and F4 are up (plain weave on the face ends 2 and 4 for this face pick), face ends F1, F3 down; of the back ends, only B2 is raised — the stitch — all other back ends down so the face pick stays above the back cloth. **Pick 4 (a back pick)**: all four F ends are marked (lifted clear); the back ends follow their own plain weave — B1 and B3 up over this back pick, B2 and B4 down. The back layer's plain weave must be phased so its ends are up over their own picks exactly as a single plain cloth's would be; checking the B columns among the b rows alone reproduces Figure 1 of Topic 1, which is the correct verification habit for any double design.

## 3. Other stitching methods and wadding

- **Centre-warp stitching:** extra ends, belonging to neither face, are woven between the layers and bind alternately into face and back — invisible from both sides; used when both faces must be flawless (fine double-faced coatings).
- **Centre-weft stitching:** extra picks do the same between the layers.
- **Wadded double cloth:** thick, soft wadding picks (or ends) are laid between two plain layers, stitched as above — quilts, winter coatings: warmth from trapped still air in the wadding, at a fraction of the weight of a solid cloth of the same thickness.
- **Unstitched forms:** stitch only at the edges and the two layers form a **tubular cloth** (sacks, hosepipes, conveyor belting); stitch in patterns and the unstitched areas inflate as **pockets** (life-jacket and air-textile structures).

## 4. Worked example — weight of a double cloth

**GSM BY LAYERS (VERIFY EACH LINE)**

Construction: arrangement 1:1; each layer 24 ends/cm and 22 picks/cm; all four systems 16.4 tex; crimps — face warp 6%, back warp 8%, face weft 5%, back weft 7%. Yarn length per m² = threads per metre × (1 + crimp).

Face warp: 2,400 × 1.06 = 2,544 m × 16.4 ÷ 1,000 = **41.7 g**

Back warp: 2,400 × 1.08 = 2,592 m × 16.4 ÷ 1,000 = **42.5 g**

Face weft: 2,200 × 1.05 = 2,310 m × 16.4 ÷ 1,000 = **37.9 g**

Back weft: 2,200 × 1.07 = 2,354 m × 16.4 ÷ 1,000 = **38.6 g**

Total = 41.7 + 42.5 + 37.9 + 38.6 ≈ **161 g/m²** — roughly two plain cloths of ~80 g/m² each, as it should be: a double cloth's weight is the sum of its layers, and its price follows the same sum.

## 5. Properties and uses

Two faces of different weave, colour or fibre (reversible coatings, blankets with a different back); **thickness and thermal insulation** from the enclosed air layer; high weight and cover without coarse yarns; the layers can be chosen independently — a fine face over a cheap, strong back. Costs: roughly double yarn consumption, slower weaving (two picks do the work of one layer's progress), and a design/drafting load that must keep four systems consistent — ends are drafted on at least 4 shaft groups (face and back separately), commonly 8+ shafts. Defects to name: **stitch marks showing on the face** (wrong stitch choice), **layer separation / bagging** (too few stitches), and **grinning** where the back colour shows through a thin face.

## Glossary

**Double cloth** — two interlaced layers woven simultaneously and stitched into one fabric. **Arrangement (ratio)** — the proportion of face to back threads, e.g. 1:1, 2:1. **Stitching** — interlacing points joining the layers (by back warp, face warp, or centre threads). **Wadding** — soft extra yarn laid between the layers. **Tubular cloth** — double cloth stitched only at the edges.

## Key points

• Four systems, one grid: face picks mark the face weave only; back picks mark all face ends up plus the back weave; stitches are extra marks.

• Figure 1's check sums (3 marks per face pick, 6 per back pick) are the quick audit of any double plain design.

• Weight = sum of the two layers (worked: ≈ 161 g/m²); choose back-warp stitching to keep the face clean.

## Practice

1. In Figure 1, state exactly which ends are up on pick 6, and classify pick 6 as face or back.
2. Why must all face ends be raised on a back pick? What defect appears in the cloth if one face end is left down?
3. A double cloth is arranged 2 face : 1 back in the warp. In a 12-end repeat, how many face and back ends are there, and what happens to the relative thickness/cover of the two layers?
4. Give two reasons for choosing back-warp stitching over face-warp stitching for a reversible coating whose face is a pale colour.

### Answer key

1. Pick 6 is a **back pick** (rows alternate f, b from pick 1 = f). Up: all four face ends (lifted clear) plus back ends B2 and B4 (the back plain weave's up-pair for this back pick) — 6 marks in all, matching the audit rule.
2. The back pick must pass *below* the entire face layer; any face end left down would be crossed by the back pick, stitching that end into the back layer at the wrong place — it appears as a wrong interlacing (a float fault) on the face and a hard stitch mark through the back.
3. 12 ends in 2:1 arrangement = **8 face ends + 4 back ends**. The face layer has twice the warp density, so (with equal yarns and weft ratio also considered) the face is the closer, heavier layer and the back a lighter support layer — the standard arrangement when the back is only a backing.
4. (i) The stitch is formed by the back end rising into the face pick, so the interlacing point is hidden under the face floats and does not break the pale face; (ii) face-warp stitching pulls a face end down through the back pick, leaving a visible dip/speck on the face — precisely what a pale face shows most.